



Lecture 6: Gang of Six

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SIST 1D#206

□ The gang of six

$$X = \frac{CP}{1+CP}FR + \frac{P}{1+CP}D - \frac{CP}{1+CP}N$$

$$Y = \frac{CP}{1+CP}FR + \frac{P}{1+CP}D + \frac{1}{1+CP}N$$

$$U = \frac{C}{1+CP}FR - \frac{CP}{1+CP}D - \frac{C}{1+CP}N$$

$$E = \frac{1}{1+CP}FR - \frac{1P}{1+CP}D - \frac{1}{1+CP}N$$

$$\begin{aligned} U &= CE = C(FR - Y) \\ &= C\left(FR - \frac{CP}{1+CP}FR - \frac{P}{1+CP}D - \frac{1}{1+CP}N\right) \\ &= \left(\frac{C}{1+CP}FR - \frac{CP}{1+CP}D - \frac{C}{1+CP}N\right) \end{aligned}$$

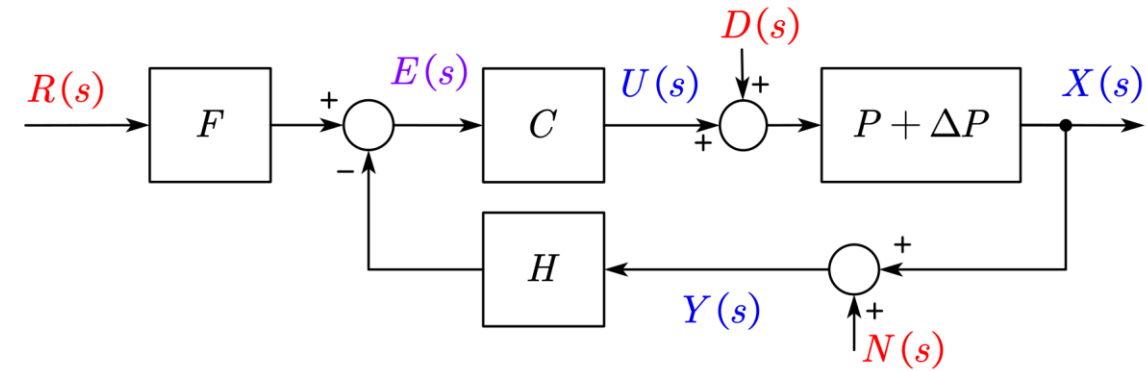


Figure 17. Closed loop system with three input channels.

$$\frac{C}{1+PC}$$

Noise sensitivity function

$$\frac{P}{1+PC}$$

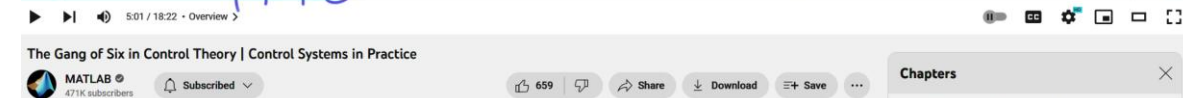
Disturbance sensitivity function

$$\frac{1}{1+PC}$$

Sensitivity function

$$\frac{PC}{1+PC}$$

Complementary sensitivity function





Sensitivity Function

- Sensitivity is the **ratio of** the fractional change in the function **to** the fractional change in the parameter **as** the fractional change of the parameter approaches zero.

$$S_{F:P} = \lim_{\Delta P \rightarrow 0} \frac{\text{Fractional change in the function, } F}{\text{Fractional change in the parameter, } P}$$

$$S_{F:P} = \frac{P}{F} \frac{\delta F}{\delta P}$$

$$= \lim_{\Delta P \rightarrow 0} \frac{\Delta F / F}{\Delta P / P}$$

$$= \lim_{\Delta P \rightarrow 0} \frac{P \Delta F}{F \Delta P}$$

Sensitivity of a Closed-Loop Transfer Function

PROBLEM: Given the system of Figure 7.19, calculate the sensitivity of the closed-loop transfer function to changes in the parameter a . How would you reduce the sensitivity?

SOLUTION: The closed-loop transfer function is

$$T(s) = \frac{K}{s^2 + as + K} \quad (7.76)$$

Using Eq. (7.75), the sensitivity is given by

$$S_{T:a} = \frac{a}{T} \frac{\delta T}{\delta a} = \frac{a}{\left(\frac{K}{s^2 + as + K} \right)} \left(\frac{-Ks}{(s^2 + as + K)^2} \right) = \frac{-as}{s^2 + as + K} \quad (7.77)$$

which is, in part, a function of the value of s . For any value of s , however, an increase in K reduces the sensitivity of the closed-loop transfer function to changes in the parameter a .

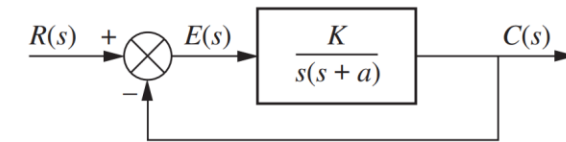


FIGURE 7.19 Feedback control system for Examples 7.10 and 7.11

$$T(s; \alpha) = \frac{N(s; \alpha)}{D(s; \alpha)}$$

$$S_{\alpha}^T = \frac{\partial \ln T}{\partial \ln \alpha} = \frac{\partial \ln N}{\partial \ln \alpha} \bigg|_{\alpha=\alpha_0} - \frac{\partial \ln D}{\partial \ln \alpha} \bigg|_{\alpha=\alpha_0}$$

- The sensitivity function of a closed loop system $T(s)$ with respect to the plant transfer function $P(s)$ can be derived as follows:

$$T(s) = \frac{CP}{1 + CP}$$

$$\Rightarrow S_P^T = \frac{P}{T} \frac{\partial T}{\partial P} = \frac{P}{\frac{CP}{1 + CP}} \frac{\partial \frac{CP}{1 + CP}}{\partial P} = \frac{1 + CP}{C} \frac{C(1 + CP) - CP(C)}{(1 + CP)^2} = \frac{1}{1 + CP}$$

- S happens to be the transfer function from noise to output.
- S is the fundamental reason why negative feedback system is superior than an open loop control system, in a sense that when the plant parameters deviate over time, the output is minimally affected.



Error Signal Analysis

(Reference tracking, disturbance rejection, and noise attenuation)

(Review) Steady State Error due to Disturbance



- Review the problem formulation of a control system two input channels: the reference signal $R(s)$ and disturbance signal $D(s)$

$$C(s) = E(s)G_1(s)G_2(s) + D(s)G_2(s)$$

$$C(s) = R(s) - E(s)$$

$$E(s) = \frac{1}{1 + G_1(s)G_2(s)}R(s) - \frac{G_2(s)}{1 + G_1(s)G_2(s)}D(s)$$

$$\begin{aligned} e(\infty) &= \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{s}{1 + G_1(s)G_2(s)}R(s) - \lim_{s \rightarrow 0} \frac{sG_2(s)}{1 + G_1(s)G_2(s)}D(s) \\ &= e_R(\infty) + e_D(\infty) \end{aligned}$$

- For a step input of $D(s) = \frac{1}{s}$ the steady state error is

$$e_D(\infty) = - \frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)}$$

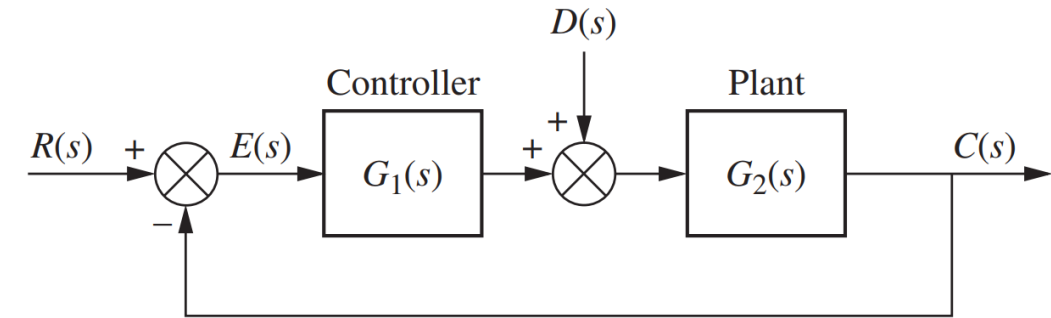


FIGURE 7.11 Feedback control system showing disturbance

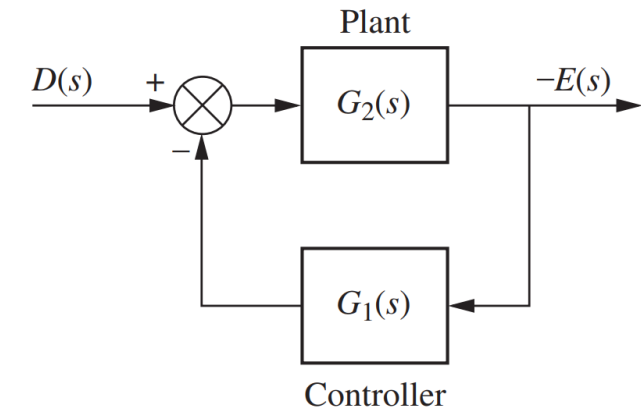


FIGURE 7.12 Figure 7.11 system rearranged to show disturbance as input and error as output, with $R(s) = 0$

Eliminating Steady State Error due to Disturbance



□ For the problem shown in the right, can you design a cascaded compensator that eliminate the steady state error due to disturbance, $e_D(\infty)$?

- The PI regulator can add an integrator in the feedback path of the closed loop system with disturbance as input and error as output, such that $G_1(0) = \infty$

$$e_D(\infty) = - \frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)}$$

PROBLEM: Find the steady-state error component due to a step disturbance for the system of Figure 7.13.

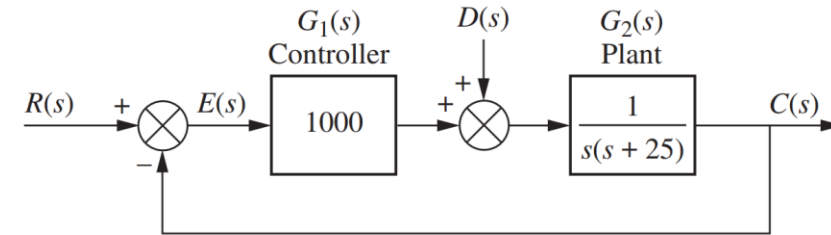


FIGURE 7.13 Feedback control system for Example 7.7

SOLUTION: The system is stable. Using Figure 7.12 and Eq. (7.62), we find

$$e_D(\infty) = - \frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)} = - \frac{1}{0 + 1000} = - \frac{1}{1000} \quad (7.63)$$

The result shows that the steady-state error produced by the step disturbance is inversely proportional to the dc gain of $G_1(s)$. The dc gain of $G_2(s)$ is infinite in this example.

- Revisit the serve control system. Four of the transfer functions are evaluated for frequency response:

$$\begin{aligned} X &= \frac{CP}{1+CP}FR + \frac{P}{1+CP}D - \frac{CP}{1+CP}N \\ Y &= \frac{CP}{1+CP}FR + \frac{P}{1+CP}D + \frac{1}{1+CP}N \\ U &= \frac{C}{1+CP}FR - \frac{CP}{1+CP}D - \frac{C}{1+CP}N \\ E &= \frac{1}{1+CP}FR - \frac{1P}{1+CP}D - \frac{1}{1+CP}N \end{aligned}$$

figure(2)

C = CVL;

P = PclosedInner * PVL;

```
subplot(141); bode(C*P/(1+C*P)); grid; h1 = findobj(gcf,'type','line'); set(h1,'linewidth',2);
subplot(142); bode(1/(1+C*P)); grid; h1 = findobj(gcf,'type','line'); set(h1,'linewidth',2);
subplot(143); bode(P/(1+C*P)); grid; h1 = findobj(gcf,'type','line'); set(h1,'linewidth',2);
subplot(144); bode(C/(1+C*P)); grid; h1 = findobj(gcf,'type','line'); set(h1,'linewidth',2);
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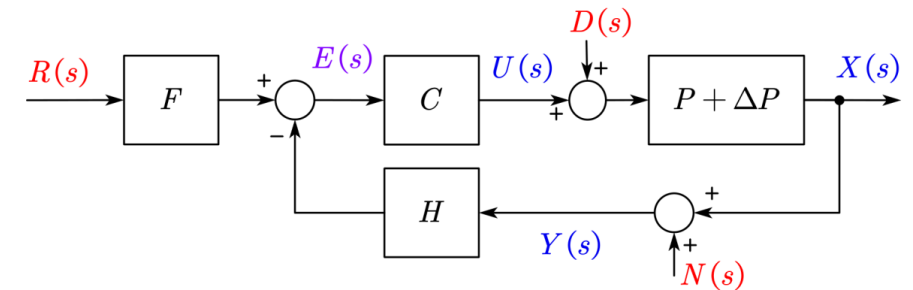
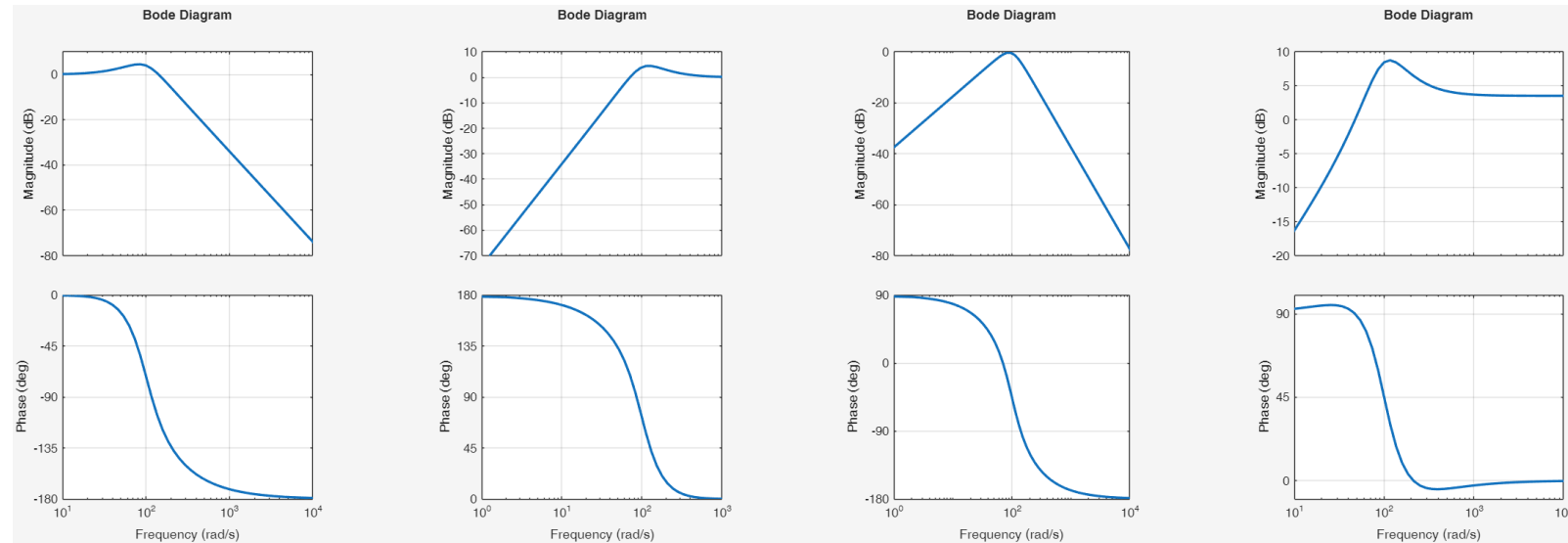


Figure 17. Closed loop system with three input channels.

Frequency Response of typical C.L. tf's



- Example paper using frequency responses to evaluate a servo control system

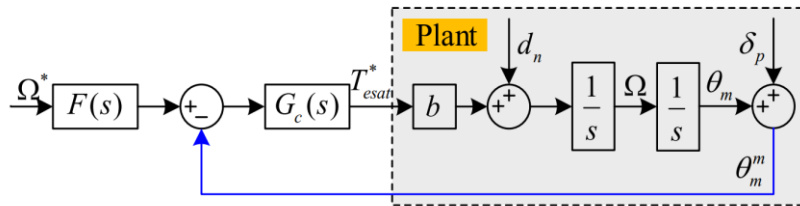


Fig. 3. Block diagram of different ADRC systems in transfer function form

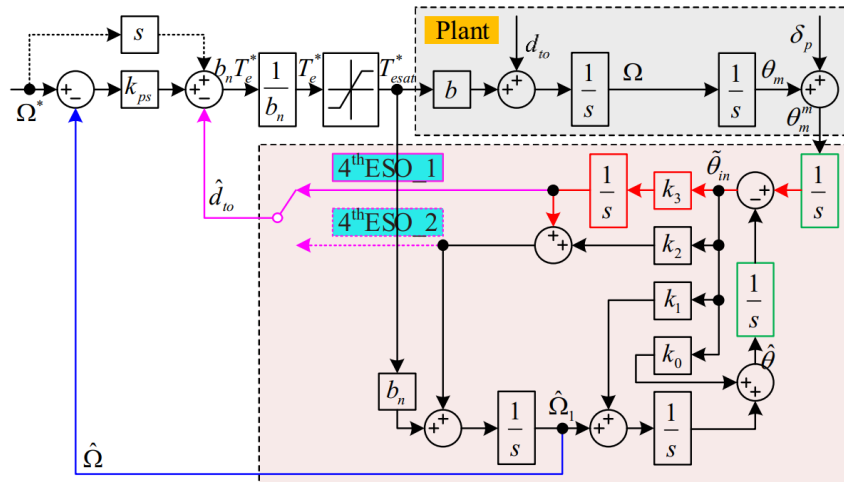


Fig. 14. The equivalent block diagram of the system 4thESO_1-2.

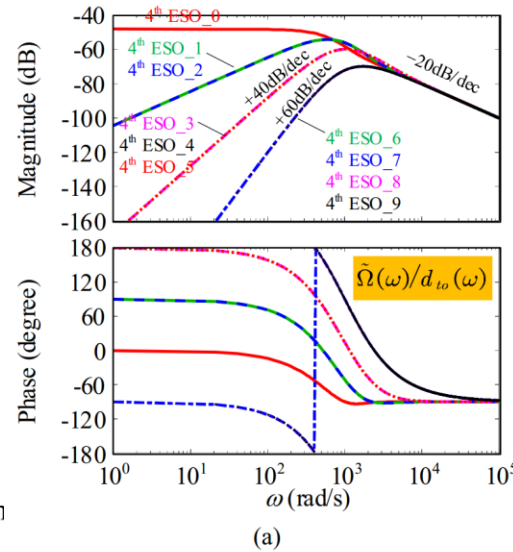


Fig. 10. Bode diagrams of the ten fourth order ESOs for observing speed.

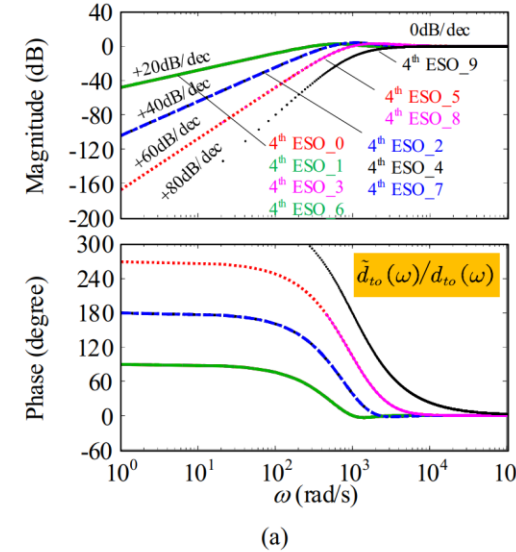


Fig. 11. Bode diagrams of the ten fourth order ESOs for observing disturbance.

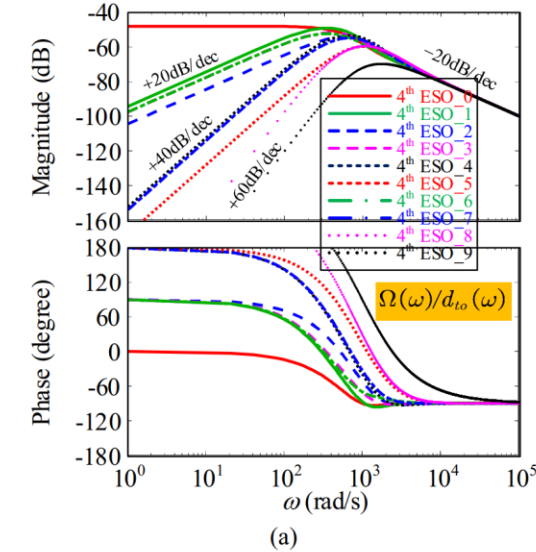
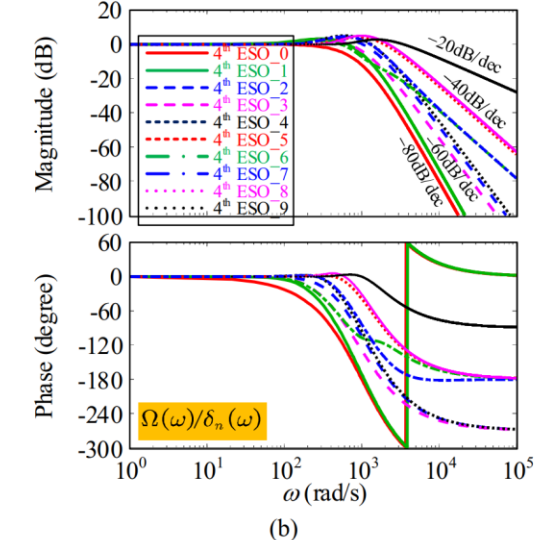
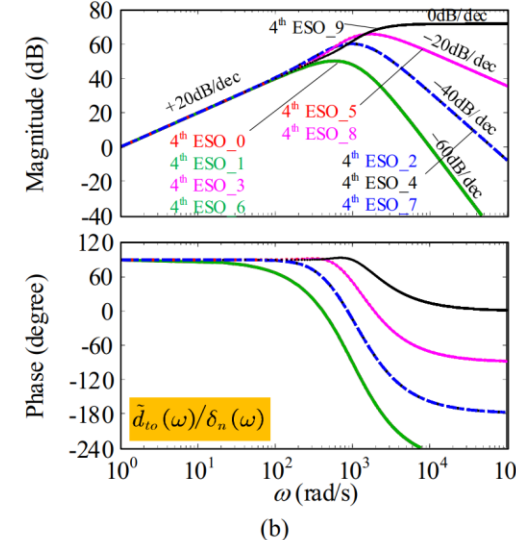
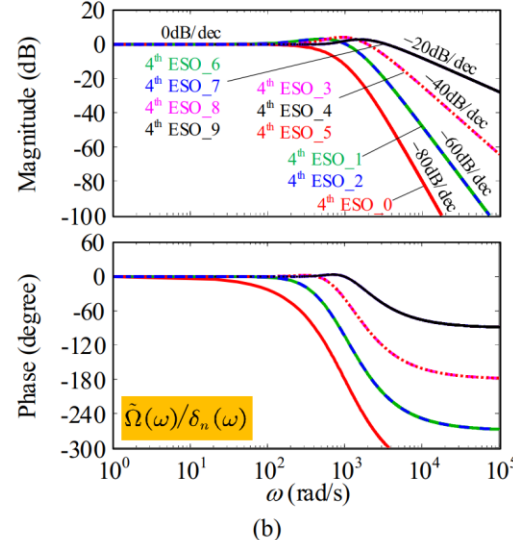


Fig. 12. Bode diagram of the ten systems under the same p_o .



Two degree of freedom PI control

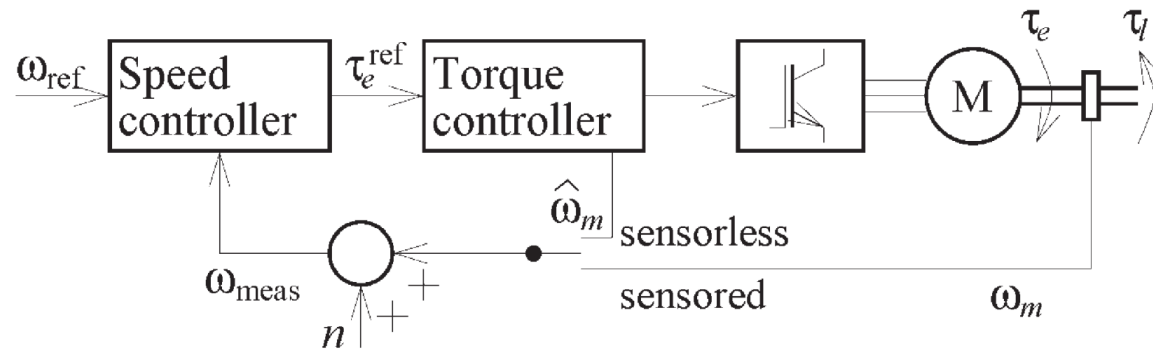


Fig. 1. Illustration of the speed control loop. The torque controller computes switching commands to the converter, which, in turn, powers the (M) motor.

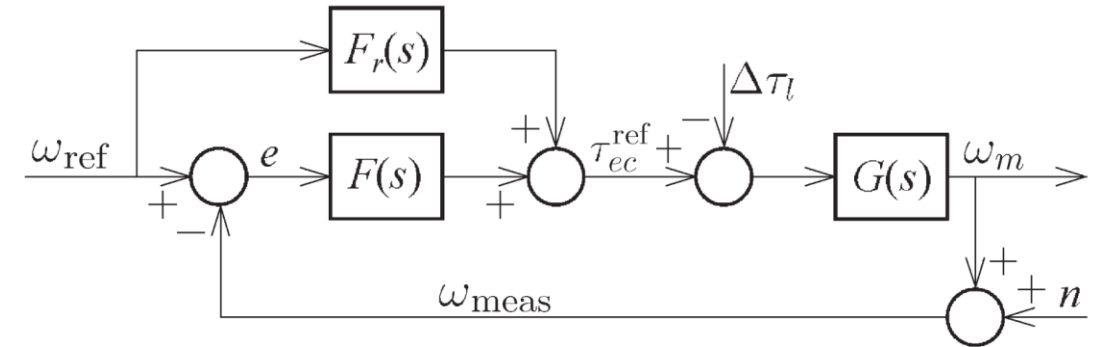


Fig. 4. Block diagram of the 2DOF structure.

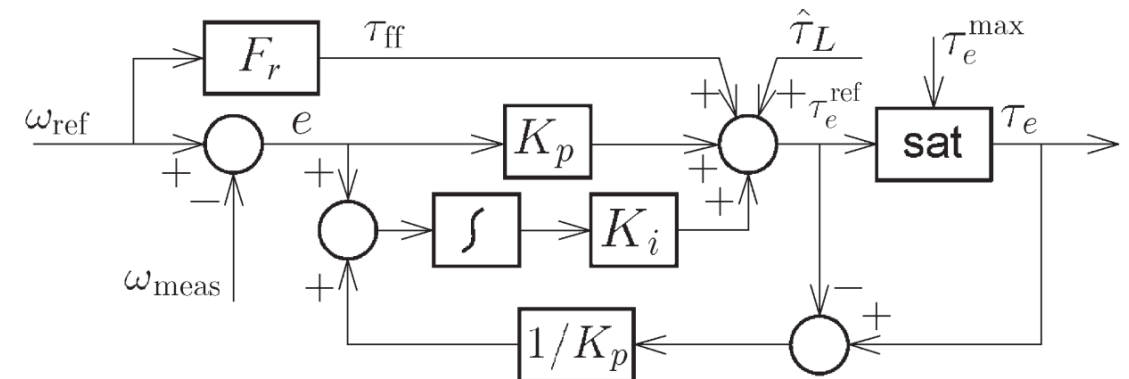


Fig. 8. Block diagram of the proposed 2DOF PI controller.